THE UNITED REPUBLIC OF TANZANIA
THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT
ANNUAL JOINT EXAMINATION

FORM THREE
BASIC MATHEMATICS

Time: 3 Hours **November 2024**

INSTRUCTIONS

1. This paper consists of sections A and section B with a total of 14 questions.
2. Attempt all questions in both sections.
3. Each question in section A carries 6 marks while each question in section B carries 10 marks.
4. Mathematical tables and non-programmable calculator may be used.
5. Write your examination number on every page of your answer sheets.

SECTION A (60 MARKS)

1. (a) i. Write in words the least or the lowest sum of the square of 1 109 and the square root of 841
   ii. Given 0.45454545... at **Seypalm** secondary school represents boys. What fraction of students at that school represents girls?
   (b) Given 0.0607879
   i. Write to 3 decimal places
   ii. Write to 3 significant figures

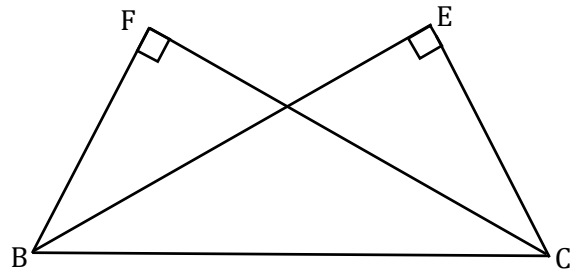
2. (a) Solve: $100^{\sqrt{x}} = \sqrt{1\,000^{x}}$
   (b) Given $M = \log_2(x+3) + \log_2(x-3)$
   i. Determine the value of x for which M is defined
   ii. Solve for x if $M = 2^2$

3. (a) Given the universal set U has 20 elements, and subsets of U are A and B, where $n(A) = 10$, $n(B) = 11$ and $n(A \cup B) = 18$, use a venn diagram to find $n(A \cap B)'$
   (b) How many elements are in a set whose subsets are 128?

4. (a) Given that: $2x - 3y + 7 = 0$
   i. Express the equation in form $y = mx + c$
   ii. Identify its gradient
   iii. Identify $y-$ intercept
   (b) The equation of a line passes through points Q (, 2) and P(4 , 4). Calculate x if the gradient of the line is 2

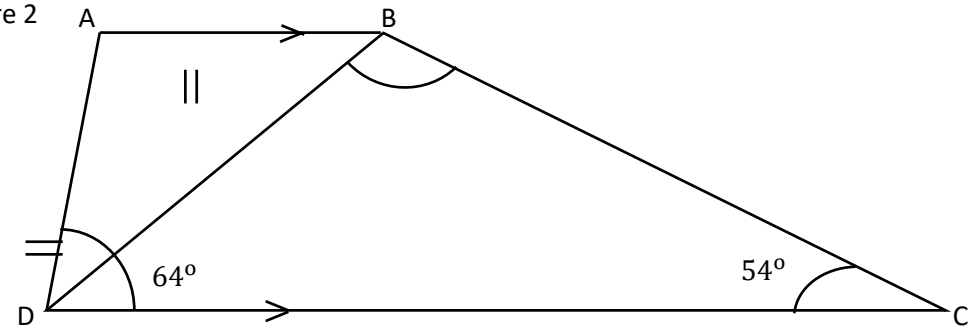
5. (a) In the following figure $\overline{BF} = \overline{CE}$,prove that $\overline{CF} = \overline{BE}$

Figure 1



(b) In the following trapezium ABCD , $\overline{AB} \parallel \overline{DC}, \overline{AD} = \overline{AB}, \angle ADC = 64^{\circ}\ and\ \angle BCD = 54^{\circ}\ , find\ \angle DBC$

Figure 2



6. (a)The weight of the mother and her daughter is 112 kg 70 g , find the weight of the mother if the daughter weighs 35 kg 59 dag

(b) In covering a distance of 30 km, Seypalm takes 2 hours more than Jonathan does. If **Seypalm** doubles his speed, then he would take 1 hour less than Jonathan would. What is Seypalm's speed?

7. (a) If $1.5x = 0.04y$, then what is the value of $\frac{y-x}{y+x}$?

(b)The following balances were extracted from the ledger of Seypalm co business on 31st January. Prepare a trial balance

Capital = 30 000 shillings, Insurance = 3 000 shilings, Cash = 18 000 shillings
Furniture = 25 000 shillings, Motor vehicle = 45 000 shillins, Discount received =7 000 shillings
Sales =68 000shillings, Discount alllowed= 4 000 shillings, Purchase= 54 000 shillings, Drawing=12 000 shillings, Creditors= 76 000 shillings, Electricity= 5 000 shillings, Debtors= 15 000shillings

8. (a) Insert 5 Arithmetic mean between 17 and 53

(b) Given that $p-2\ , p-1\ , and\ 3p-5$ are 3 consecutive terms of G.P. Find the possible values of p

9. (a) If $\text{Sin}x = \frac{2y}{4y}$, find the following:

i. the value of x

ii. tan x given that x is an acute angle

(b) Calculate the length of a diagonal of a square with sides 9 cm. Leave the answer in a surd form.

10. (a) i. Without using calculator, find the value of $\frac{(67.542)^2-(32.458)^2}{75.458-40.374}$. Write the answer in scientific notation

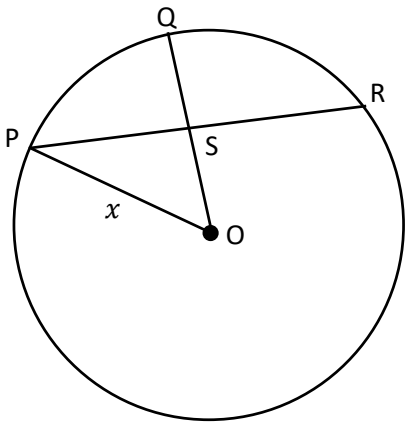
ii. Solve the following equation: $\frac{2x}{1+\frac{1}{1+\frac{x}{1-x}}} = 1$

(b) Solve: $\frac{2P-1}{2} = \frac{2P+2}{P+1}$

SECTION B (40 MARKS)

11. (a)In the following circle with centre O ,$OQ \perp PR, PR = 12\ units$ and $SQ = 2units$. Determine x

Figure 3



(b) The following table shows the height of a sample of 20 Seypalm factory workers

Height (cm)	125 − 130	131 − 136	137 − 142	143 − 148	149 − 154
Cumulative frequency	5	13	25	29	30

i. Find the mean height of the workers

ii. What is the mode height of the workers?

iii. Draw a cumulative frequency curve and use it to estimate the median height

12. (a) The speed of a boat travelling from point A $(6^\circ N, 33^\circ E)$ to point B $(6^\circ N, 39^\circ E)$ is 20 knots. After how many hours will it reach at point B?

(b) Let A$\{-2, -1, 0, 1, 2\}$ and B$\{0, 1, 2, 3, 4\}$. Draw an arrow diagram to illustrate the relation, which connects each element of A with its square.

13. (a) Find the image of the point Q (3 , 2) after rotating through270° clockwise about the origin

(b) In how many years would **Seypalm's investment** double, if 5 000 000 shillings is invested at 5.5% interest compounded semi- annually?

14. (a) Given $f(x) = \begin{cases} -3\ when\ x < -2 \\ 1\ when -2 \leq x < 1 \\ 4\ when\ x > 1 \end{cases}$

i. Sketch the graph of $f(x)$

ii. State the domain and range $f(x)$

iii. Is $f(x)$ a one− to - one function?

(b) Draw the graph of $f(x) = -x^2 + 2$ and find its domain and range.